away. Pick fights with people smaller than you are.' Things are a bit more complicated if the importance of size is less certain. If large size confers only a slight advantage, the strategy I have just mentioned is still stable. But if the risk of injury is serious there may also be a second, 'paradoxical strategy'. This is: 'Pick fights with people larger than you are and run away from people smaller than you are'! It is obvious why this is called paradoxical. It seems completely counter to common sense. The reason it can be stable is this. In a population consisting entirely of paradoxical strategists, nobody ever gets hurt. This is because in every contest one of the participants, the larger, always runs away. A mutant of average size who plays the 'sensible' strategy of picking on smaller opponents is involved in a seriously escalated fight with half the people he meets. This is because, if he meets somebody smaller than him, he attacks; the smaller individual fights back fiercely, because he is playing paradoxical; although the sensible strategist is more likely to win than the paradoxical one, he still runs a substantial risk of losing and of being seriously injured. Since the majority of the population are paradoxical, a sensible strategist is more likely to be injured than any single paradoxical strategist.

Even though a paradoxical strategy can be stable, it is probably only of academic interest. Paradoxical fighters will only have a higher average pay-off if they very heavily out-number sensible ones. It is hard to imagine how this state of affairs could ever arise in the first place. Even if it did, the ratio of sensibles to paradoxicals in the population only has to drift a little way towards the sensible side before reaching the 'zone of attraction' of the other ESS, the sensible one. The zone of attraction is the set of population ratios at which, in this case, sensible strategists have the advantage: once a population reaches this zone, it will be sucked inevitably towards the sensible stable point. It would be exciting to find an example of a paradoxical ESS in nature, but I doubt if we can really hope to do so. (I spoke too soon. After I had written this last sentence, Professor Maynard Smith called my attention to the following description of the behaviour of the Mexican social spider, *Oecobius civitas*, by J. W. Burgess: 'If a spider is disturbed and driven out of its retreat, it darts across the rock and, in the absence of a vacant crevice to hide in, may seek refuge in the hiding place of another spider of the same species. If the other spider is in residence when the intruder enters, it does not attack but darts out and seeks a new refuge of its own.